DEVOPS GIT BASICS PROJECT REPORT

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Project Title: Git Basics – Repository Creation and Version Control
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1. Introduction

This project demonstrates the implementation of basic Git and GitHub operations. It includes initializing a local repository, creating and tracking files, committing changes, connecting to a remote GitHub repository, and managing version updates.
The purpose of this project is to understand version control workflows used in DevOps practices.

2. Project Objective

The objectives of this project are:

To create a public GitHub repository.
To initialize a local Git repository.
To create and track project files.
To commit changes using Git.
To connect a local repository to GitHub.
To push code to a remote repository.
To update project version and maintain version control.

3. Repository Creation on GitHub

A public GitHub repository named:

devops-git-basics

was created without initializing it with README, .gitignore, or license files.

```
PS C:\Users\user\Documents\git repository> git init
Initialized empty Git repository in C:/Users/user/Documents/git reposi
tory/.git/
PS C:\Users\user\Documents\git repository> echo "I am a responsible an
d motivated person. I enjoy learning new skills and improving myself b
oth personally and professionally." > README. md
PS C:\Users\user\Documents\git repository> echo " v1.0.0 " > version.t
xt
PS C:\Users\user\Documents\git repository> git add .
PS C:\Users\user\Documents\git repository> git commit -m " comitting R
EAD. me and version.txt "
[master (root-commit) 68283e7]  comitting READ. me and version.txt
 2 files changed, 3 insertions(+)
 create mode 100644 README
 create mode 100644 version.txt
PS C:\Users\user\Documents\git repository> git branch -M main
PS C:\Users\user\Documents\git repository> git branch
* main
PS C:\Users\user\Documents\git repository> git remote add origin https
://github.com/shariquafarooque-sha/devops-git-basics.git
PS C:\Users\user\Documents\git repository> git push -u origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (3/3), done.
Writing objects: 100% (4/4), 380 bytes | 380.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shariquafarooque-sha/devops-git-basics.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\user\Documents\git repository> echo " v1.0.1 " > version.t
xt
PS C:\Users\user\Documents\git repository> git add version.txt
```

4. Local Repository Setup

4.1 Initialize Git Repository

Command Used:

git init

Output:

Initialized empty Git repository in C:/Users/user/Documents/git repository/.git/

Explanation:

The `git init` command creates a hidden `.git` folder which enables Git to track changes in the project directory.

5. File Creation

5.1 Creating README.md

Command Used:

echo "I am a responsible and motivated person. I enjoy learning new skills and improving myself both personally and professionally." > README.md

Purpose:

The README.md file contains a short personal bio and describes the repository.

5.2 Creating version.txt

Command Used:

echo "v1.0.0" > version.txt

Initial Version:

v1.0.0

The version.txt file is used to maintain project version information.

6. Staging and First Commit

6.1 Adding Files to Staging Area

Command:

git add .

6.2 Committing Files

Command:

git commit -m "committing README and version.txt"

Output:

```
[master (root-commit) 68283e7] committing README and version.txt
2 files changed, 3 insertions(+)
create mode 100644 README.md
create mode 100644 version.txt
```

Explanation:

This was the first commit (root commit). Both files were successfully added to version control.

7. Renaming Branch to Main

Command:

```
git branch -M main
```

Verification:

```
git branch
```

Output:

```
main
```

Explanation:

The branch was renamed to "main" because GitHub uses main as the default branch.

8. Connecting to Remote Repository

Command:

```
git remote add origin https://github.com/shariquafaroque-sha/devops-git-basics.git
```

Explanation:

The local repository was connected to the remote GitHub repository using the remote name "origin".

9. First Push to GitHub

Command:

git push -u origin main

Output:

Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Writing objects: 100% (4/4), done.
[new branch] main -> main
branch 'main' set up to track 'origin/main'.

Explanation:

The local repository was successfully pushed to GitHub. The -u option set upstream tracking.

10. Updating Project Version

```
To https://github.com/shariquafarooque-sha/devops-git-basics.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
PS C:\Users\user\Documents\git repository> echo " v1.0.1 " > version.t
xt
PS C:\Users\user\Documents\git repository> git add version.txt
PS C:\Users\user\Documents\git repository> git commit -m "updated the
version of the application"
[main ab47bc0] updated the version of the application
 1 file changed, 1 insertion(+), 1 deletion(-)
PS C:\Users\user\Documents\git repository> git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 301 bytes | 301.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/shariquafarooque-sha/devops-git-basics.git
   68283e7..ab47bc0  main -> main
PS C:\Users\user\Documents\git repository>
```

10.1 Updating version.txt

Command:

echo "v1.0.1" > version.txt

Old Version:

v1.0.0

New Version:

v1.0.1

10.2 Staging the Updated File

Command:

```
git add version.txt
```

10.3 Committing the Update

Command:

```
git commit -m "updated the version of the application"
```

Output:

```
[main ab47bc0] updated the version of the application
1 file changed, 1 insertion(+), 1 deletion(-)
```

Explanation:

The old version line was replaced with the new version.

10.4 Pushing Updated Version

Command:

```
git push origin main
```

Output:

```
Enumerating objects: 5, done.
Writing objects: 100% (3/3), done.

main -> main
```

Explanation:

The updated version was successfully pushed to GitHub.

11. Final Repository Link

https://github.com/shariquafaroque-sha/devops-git-basics

12. Conclusion

This project successfully demonstrated the practical use of Git and GitHub in a DevOps workflow. The tasks completed include:

Repository initialization

File tracking
Making commits
Branch management
Remote repository connection
Version updating
Pushing changes to GitHub

The project provided hands-on experience with essential Git commands used in real-world DevOps environments.